

Localization of Eye Region in Infrared Thermal Images using Deep Neural Network

Madura Meenakshi R.¹, Padmapriya N.², Venkateswaran N.³, Ravikumar R.⁴, and Ramya Chelliah⁵

^{1,2,3}Department of Mathematics, Sri Sivasubramaniya Nadar College of Engineering,
Kalavakkam, Tamil Nadu, India

^{4,5}Department of Ophthalmology, Tagore Medical College Hospital, Chennai, Tamil Nadu, India

¹madurameenakshir@ssn.edu.in, ¹padmapriyan@ssn.edu.in, ³venkateswarann@ssn.edu.in,

⁴nrravikumar@gmail.com, ⁵ramche12@yahoo.co.in

Abstract—Infrared (IR) thermal images are produced by recording the thermal radiation emitted from the object. It has been increasingly used in industrial installations, surveillance, evaluating human health, etc. Medical IR thermal images are used as a tool for early diagnosis of certain diseases in the human beings. It can detect the heat patterns of the body as well as irregular blood flow. Compared to visual images, thermal images have low resolution and low contrast. Hence in IR thermal images eye detection is more challenging, for ocular surface disease diagnosis. This article proposes an automatic eye localization method from IR thermal images using YOLO v2 object detector. The performance of eye localization in IR thermal images using YOLO v2 shows mean average precision 97% and mean intersection over union (IoU) 90% for test images.

Index Terms—Infrared thermal image, Region of interest, Eye localization, YOLO v2, Deep neural network.

I. INTRODUCTION

Thermography and image processing are useful tools because they facilitate a noninvasive technique and fast diagnosis. Ocular thermography as a diagnostic tool detects ocular blood flow in eye diseases [1]. An accurate localization of eye will help in the diagnosis of ocular diseases efficiently. Compared to visual images, IR thermograms do not possess clear boundaries which makes eye detection challenging. So far, there are few techniques reported on the automated localization of eye and cornea. J.H. Tan et al. [2,3] proposed an automatic localization of eye and cornea using snake algorithm and target tracing function for initialization of snake coupled with genetic algorithm. Shangfei Wang et al. [4] followed an automatic eye localization method from long wave infrared thermal images both with eyeglasses and without eyeglasses. Mariusz et al. [5] adopted an eye localization algorithm which consist of two stages. The first part of the algorithm is a block for generating appropriated features describing eye areas. The second stage is a decision block based on a neural network allowing for effective classification of pre-designated areas.

Mateusz et al. [6] presented eye detection in thermal images using the virtual high dynamic range technique, to enhance performance of the dense grid of scale-invariant feature descriptors, combined with the bag-of-visual-words approach. Traditional methods such as SIFT descriptor, Haar like features and other feature description methods usually

use handcrafted features to detect eyes from infrared thermal images directly. In recent years, region proposal CNN architectures have achieved incredible success on object detection [7–9] as well as eye detection tasks in visual images [10]. J. Huang et al. [11] guides in selection of detection architect which has a high speed/accuracy trade-off for various visual image datasets. However, when applying region proposal CNN architectures to eye detection in IR thermal images is still very challenging. Since infrared thermal images has low resolution, a fast and accurate network is a need for eye localization. The main objective of this research is to develop a framework for eye localization which increases the speed and accuracy of the detection of eyes in infrared thermal images. YOLO v2 [12] achieves very good performance in standard low resolution visual image datasets. Thus, a YOLO v2 based eye localization in infrared thermal images is proposed in this article. The contribution of the paper are as follows

- Automatic eye localization for ocular surface disease diagnosis
- Fast eye localization in IR thermal eye images.
- YOLO v2 based network accurately locates eye region with high average precision and with low average false prediction rate

The paper is organized as follows. Section II presents the detailed description of our proposed eye detection method. Section III presents a detailed discussion on the results. Section IV concludes the paper.

II. PROPOSED METHODOLOGY

A. Image Acquisition Process

The data set was created from the scratch using a bench top FLIR A35sc thermal imaging camera with standard laws and the Dell Latitude Laptop (5th Gen/ 8GB/ 1TB with windows 8) which was connected to PoE injector 16 W (with multi-plugs) of the camera through ethernet cables. The experiment was carried out in a room with all the windows and doors kept closed [13]. The prerequisites for the acquisition are presented in Table I.

The image acquisition took was carried out at department of Ophthalmology, Tagore Medical College Hospital, Rathinamangalam, Chennai and SSN College of Engineering,

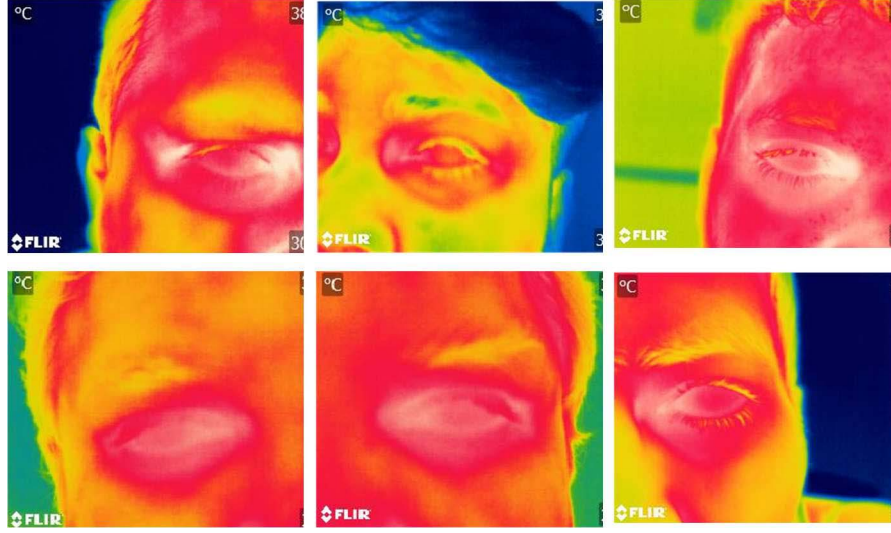


Fig. 1. Sample Infrared Thermal images of eye.

TABLE I
PREREQUISITES FOR THE IMAGE ACQUISITION.

S. No	Parameters	Values
1.	Atmospheric Temperature	24°C to 26°C
2.	Reflexive Temperature	25°C to 26°C
3.	Relative humidity	65% to 75%
4.	Emissivity	0.98
5.	Distance between eye and Camera.	0.2 m

Kalavakkam, Tamil Nadu. Ethical approval for this study has been obtained from Human Ethical Committee of SSN Institutions, Kalavakkam, Tamil Nadu. The dataset consists of 2003 IR thermal eye images. Sample images from the acquired dataset are given in Fig. 1.

B. Architecture of YOLO v2

YOLO v2 proposed in [11] divides the input image into an $S \times S$ grid. Each grid cell detects the object if the center of the object exists in that cell. Each cell predicts the bounding boxes and their confidence scores. The confidence scores are the probability of predicted box containing an object and the accuracy of its overlap with ground truth box. The intersection over union is calculated by the predicted box and the ground truth. Thus,

$$\text{confidence Score} = P_r(\text{Obj}) \times \text{IoU} \quad (1)$$

$$\text{IoU} = \frac{\text{Area of } B_1 \cap \text{Area of } B_2}{\text{Area of } B_1 \cup \text{Area of } B_2} \quad (2)$$

In addition to confidence score the model also outputs the location and the dimensions of the bounding boxes as (x, y, w, h) where (x, y) represents coordinates of the center of the box relative to the bounds of the grid cell, w and h are the width and height of the predicted box relative to the whole image.

Limitations of YOLO such as localization error and low recall than region proposal methods were improved in YOLO v2 by batch normalization, anchor boxes and high-resolution classifier.

C. Proposed Modified Network Architecture

YOLOv2 network adopted in proposed work is a composition of feature extraction network followed by detection network. A pretrained CNN network, ResNet50 [14], which is trained on ImageNet database [15] is taken as the base network for feature extraction. ResNet50 has 50 deep layers in that 48 are convolutional layers, one max pooling layer, one average pooling layer followed by a fully connected layer.

The network has an image input size of 224-by-224 with 3 channels. Any layer other than fully connected layer can be taken as feature extraction layer. In this proposed model, activation relu-40 is considered as the feature extraction layer. The output size of the feature map is 14-by-14. YOLO v2 based detection network follows after the feature extraction. The fully connected layer is removed from the base network and YOLO adds three 3×3 convolutional layers followed by 1×1 convolutional layer. In order to regularize the data YOLO uses batch normalization. Batch normalization layer works as feature scaling and it happens after convolution layer and before activation layer. The block diagram of the proposed work is given in Fig. 2.

YOLO v2 uses anchor boxes for location prediction instead of region proposals. The anchor box was estimated by the ground truth bounding boxes. The distribution of the ground truth box size is presented in Fig. 3. Using K-means clustering algorithm similar boxes were grouped by IoU distance metric. Since IoU is invariant to the size of the boxes, this metric is used for the estimation of anchor boxes. The number of anchor boxes is another parameter which improves the training of the model. For our training model the number of anchor boxes taken as 7.

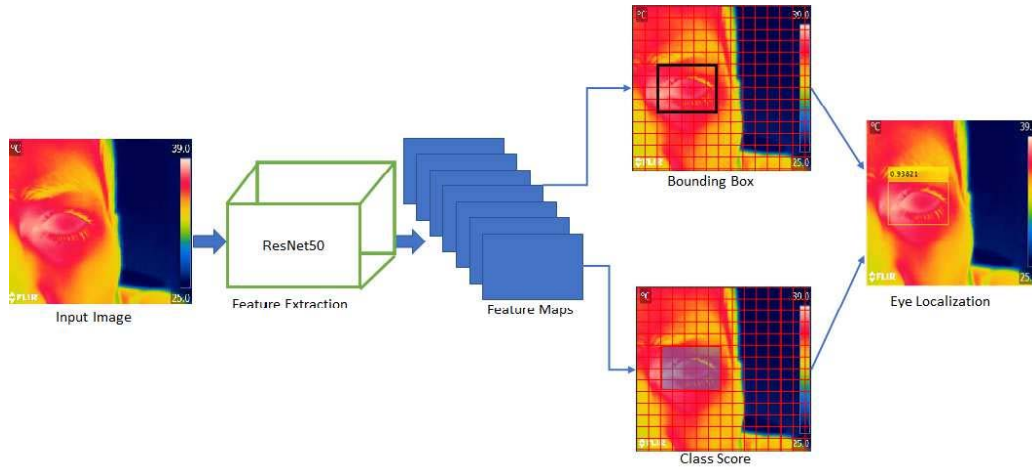


Fig. 2. Block diagram of the proposed eye localization using YOLO v2.

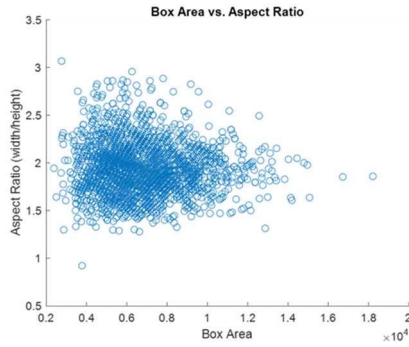


Fig. 3. The range of object sizes presented in the data set.

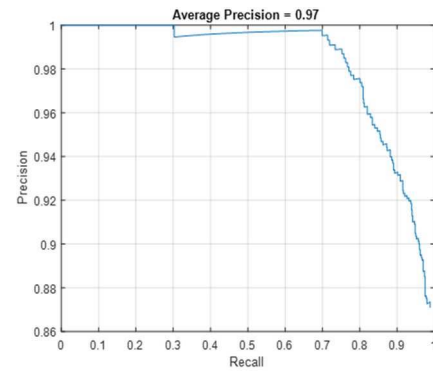


Fig. 4. The precision/recall curve.

III. RESULTS AND DISCUSSION

A. Network Training for Classification and Detection

For training the model, the labeling of 2003 infrared thermal eye images which is acquired using FLIR A35 camera was done by Image Labeler app of Matlab 2020b (Mathworks, India). Out of this 60% of the images were taken for training, 10% were taken for validation and the remaining 30% were considered for testing.

YOLO divides the input images into grids cells. Here, the output size of the feature map 14-by-14 is taken as the grid cell size. The anchor boxes are used to locate the objects in each grid cells. At first YOLO pretrains the classifier on ImageNet using 224×224 resolution inputs, then increases the resolution to 448×448 for detection, it requires the network time to simultaneously adjust to the new resolution input and perform the detection tasks. YOLO trains the network for 20 epochs using stochastic gradient descent with initial learning rate 10^{-3} and momentum of 0.9. Training the detector for the created dataset takes 20 minutes.

B. Evaluation of the Network for Test Images

The performance of YOLO v2 is evaluated by test set. Eye localization shows good performance in infrared thermal images with mean average precision 97% and mean *IoU* as

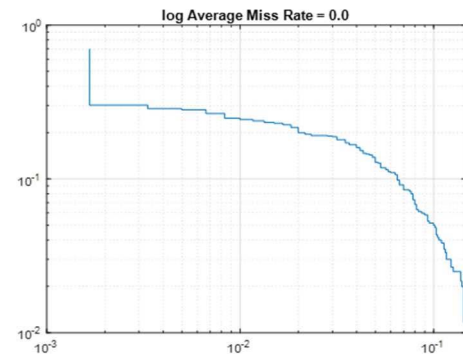


Fig. 5. Average miss rate curve.

90%. Fig. 4 shows the detector at varying levels of recall and Fig. 5 shows the log average miss rate over false positive per image of the detector.

The performance of the detector for the sample images in Fig. 3 are displayed in Fig. 6 by the predicted bounding box and confidence score. The proposed model detects the eye region in 0.692 seconds (approx.) for real time infrared thermal eye images.

The performance of the detector is compared with fast RCNN and faster RCNN based on the metrics such as

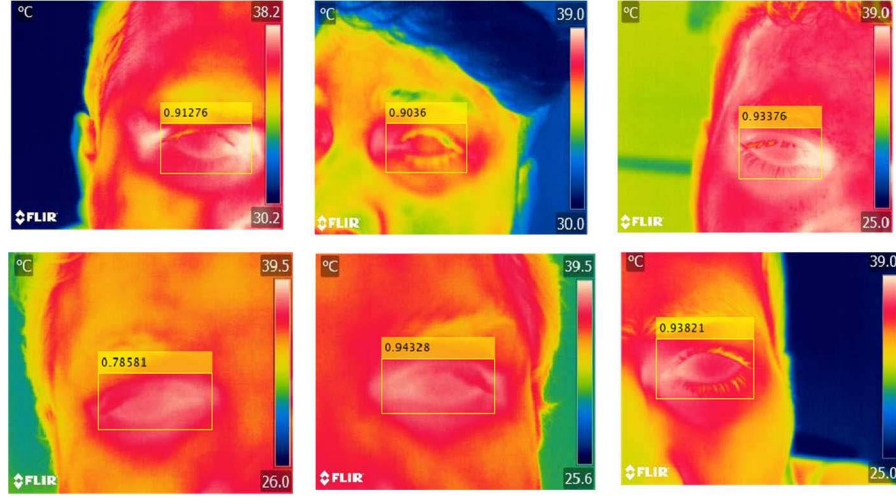


Fig. 6. Images with predicted bounding box and confidence score.

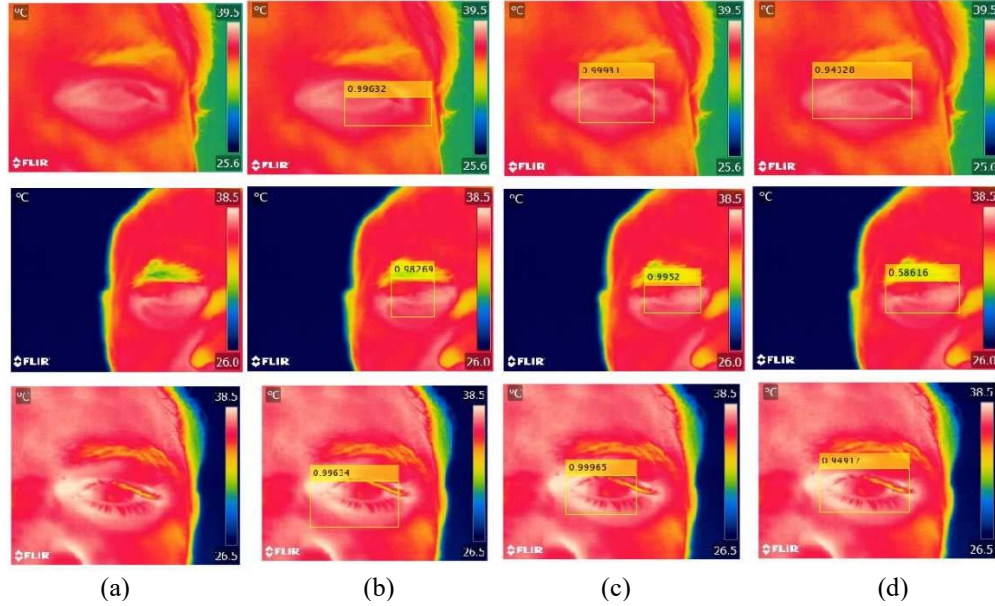


Fig. 7. Detection results. (a) sample images, (b) eye detection performed by Fast RCNN, (c) eye detection performed by Faster RCNN, (d) eye detection performed by YOLOv2.

Intersection over Union (IoU), Average Precision (AP), and mean average precision (MAP).

Average precision is a measure that combines recall and precision for ranked retrieval results. For one information need, the average precision is the mean of the precision scores after each predicted region is retrieved [16].

$$\text{Average Precision (AP)} = \frac{\sum_r P_r}{R}. \quad (3)$$

where, P and R are the precision and recall respectively and r denotes the rank.

The *Mean Average Precision* (MAP) [17] is the calculated as arithmetic mean of the average precision values for the predicted region over a set of n precision values. Here n is taken as 55. It can be expressed as follows:

$$\text{MAP} = \frac{1}{4} \sum_n AP_n \quad (4)$$

where, AP_n is the average normalised precision.

The evaluation performance of the YOLO v2, fast RCNN and faster RCNN are presented in Table II. The pretrained CNN model, Resnet50 which is trained on ImageNet database [15] is considered as the base network in all the models discussed in this paper.

The simulation results of the detection models are displayed in Fig. 7. It is clear that all the models predict the eye region with high confidence score. Though the confidence scores are high, YOLO v2 detector has drawn the predicted bounding box covering the maximum eye region. For ocular disease diagnosis, precise localization of eye region is more important than general eye recognition. Therefore, YOLO v2 out performs other models in accurate eye localization on

TABLE II
DETECTION RESULTS ON THE CREATED DATASET FOR DETECTION MODELS.

Metrics	Detection Model		
	Fast RCNN	Faster RCNN	YOLO v2
Intersection over Union (IOU)	0.5467	0.4313	0.6898
Average Precision (AP)	0.1029	0.8183	0.9561
Mean Average Precision (MAP)	0.4530	0.8398	0.9807

infrared thermal images.

IV. CONCLUSION

Early detection of ocular diseases can be done using infrared ocular thermal images. Initially locating the eye region in infrared thermal images are crucial for the diagnosis. Though there are many handcrafted methods exists, YOLO v2 detector supports the automatic localization of eye region with high accuracy and high speed. This paper presents the detection process of YOLO v2 network for the localization of eye region.

ACKNOWLEDGMENT

The authors like to thank Dr. V. N. Sheela Ramani, Dr. G. Sruthi and the staffs of the department of Ophthalmology and the management of Tagore Medical College, Rathinamangalam, Chennai, for their continuous support for this study. The authors like to thank the management of Sri Sivasubramania Nadar College of Engineering, Kalavakkam, Tamil Nadu, for providing the camera and other facilities to carry out this work.

REFERENCES

- [1] A. Sodi, B. Giambene, P. Miranda, G. Falaschi, A. Corvi, and U. Menchini, "Ocular surface temperature in diabetic retinopathy: A pilot study by infrared thermography," *European Journal of Ophthalmology*, vol. 19, no. 6, pp. 1004–1008, 2009. <https://doi.org/10.1177/112067210901900617>
- [2] J. H. Tan, E. Y. K. Ng, and U. R. Acharya, "Detection of eye and cornea on IR thermogram using genetic snake algorithm," *9th International Conference on Quantitative Infrared Thermography*, Krakow, Poland, 2008, pp. 143–150.
- [3] J.-H. Tan, E. Y. K. Ng, and U. Rajendra Acharya, "Automated detection of eye and cornea on infrared thermogram using snake and target tracing function coupled with genetic algorithm," *Quantitative InfraRed Thermography Journal*, vol. 6, no. 1, pp. 21–36, 2009, DOI: 10.3166/qirt.6.21-36
- [4] S. Wang, Z. Liu, P. Shen, and Q. Ji, "Eye localization from thermal infrared images," *Pattern Recognition*, vol. 46, pp. 2613–2621, 2013. <http://dx.doi.org/10.1016/j.patcog.2013.03.001>.
- [5] M. Marzec, A. Lamza, Z. Wróbel, and A. Dziech, "Fast eye localization from thermal images using neural networks," *Multimed Tools Applications*, 2016. DOI 10.1007/s11042-016-4094-7.
- [6] M. Knapik and B. Cyganek, "Fast eyes detection in thermal images," *Multimedia Tools and Applications*, 2020. <https://doi.org/10.1007/s11042-020-09403-6>.
- [7] R. Girshick, et al. "Rich feature hierarchies for accurate object detection and semantic segmentation," *2014 IEEE Conference on Computer Vision and Pattern Recognition*, 2014.
- [8] R. Girshick, "Fast r-cnn," *Proc. IEEE International Conference on Computer Vision*, 2015.
- [9] S. Ren, K. He, R. Girshick, and J. Sun, "Faster R-CNN: Towards real-time object detection with region proposal networks," *Advances in Neural Information Processing Systems*, vol. 28, 2015.
- [10] J. Cui, F. Chen, D. Shi, and L. Liu, "Eye detection with faster R-CNN," in *Proc. International Conference on Advances in Computer Technology, Information Science and Communications (CTISC 2019)*, pp. 111–116. DOI: 10.5220/0008096201110116.
- [11] J. Huang, V. Rathod, C. Sun et al., "Speed/accuracy trade-offs for modern convolutional object detectors," 2017, <https://arxiv.org/abs/1611.10012>.
- [12] J. Redmon, and A. Farhadi, "YOLO9000: Better, faster, stronger," *2017 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*. IEEE, 2017.
- [13] R. D. Freeman and I. Fatt, "Environmental influences on ocular temperature," *Investigative Ophthalmology & Visual Science*, vol. 12, pp. 596–602, 1973.
- [14] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conference on Computer Vision and Pattern Recognition*, pp. 770–778, 2016.
- [15] *ImageNet*. <http://www.image-net.org>
- [16] E. Zhang and Y. Zhang, "Average Precision," in: LIU L., ÖZSU M.T. (eds) *Encyclopedia of Database Systems*. Springer, Boston, MA, 2009. https://doi.org/10.1007/978-0-387-39940-9_482
- [17] S. M. Beitzel, E. C. Jensen, and O. Frieder, "MAP," in: LIU L., ÖZSU M.T. (eds) *Encyclopedia of Database Systems*. Springer, Boston, MA, 2009. https://doi.org/10.1007/978-0-387-39940-9_492